

Refund Flow

Gema Event Management Platform
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1. Refund Policy

Component	Refundable?	Notes
Ticket price (subtotal – coupon)	Yes	Full amount refunded
Service fee	No	Platform revenue, non-refundable
Tax	No	Non-refundable
Coupon discount	N/A	Already deducted from subtotal

Condition: `order.paymentStatus === 'paid'` required. Any other status returns `refundAmount: 0`.

2. Refund Amount Calculation

```
flowchart TD
    A[Order] --> B{paymentStatus == 'paid'?}
    B -->|No| C[Return: refundAmount=0, everything=0]
    B -->|Yes| D[serviceFee = order.serviceFee or 0]
    D --> E[tax = order.tax or 0]
    E --> F[ticketPrice = order.subtotal - couponDiscount]
    F --> G[nonRefundableAmount = serviceFee + tax]
    G --> H[refundAmount = Math.max 0, ticketPrice]
    H --> I[Return RefundCalculation]
```

3. Single Order Refund

```
sequenceDiagram
    participant Initiator as User/Admin
    participant RS as RefundService.processRefund()
    participant Mongo as MongoDB (session)
    participant PS as PaymentService
    participant Stripe as Stripe API

    Initiator->>RS: processRefund({orderId, reason, initiatedBy, cancellationType})
    RS->>Mongo: startSession() → startTransaction()
    RS->>Mongo: Order.findById(orderId).session(session)
    Mongo-->>RS: 404 Order not found → abort
    RS->>RS: calculateRefundableAmount(order, cancellationType)
    alt refundAmount == 0
        RS-->>Initiator: {success:false, refundAmount:0}
    end
    end
    RS->>PS: resolve correct Stripe instance
    Note over PS: Vendor Stripe OR Platform Stripe\nbased on payment routing used at time of purchase
    RS->>Stripe: refunds.create({payment_intent, amount: refundAmountInFils})
    Stripe-->>RS: Stripe refund error → abort transaction
    RS->>Mongo: Order.findByIdAndUpdate({paymentStatus:'refunded', refundId, refundedAt})
    RS->>Mongo: CancellationLog.create({orderId, eventId, userId, reason, refundAmount, initiatedBy, cancellationType})
    RS->>Mongo: commitTransaction()
    RS-->>Initiator: RefundResult {success:true, refundId, refundAmount, nonRefundableAmount, serviceFee, tax}
```

4. Batch Refund (Event Cancellation)

```
sequenceDiagram
    participant Admin as Admin
    participant RS as RefundService.processEventRefunds()
    participant DB as MongoDB
    participant Stripe as Stripe

    Admin->>RS: processEventRefunds(eventId, reason)
    RS->>DB: Order.find({eventId, paymentStatus:'paid'})
    DB-->>RS: array of paid orders
    RS->>RS: Initialize BatchRefundResult counters
    loop For each order
        RS->>RS: processRefund({orderId, cancellationType:'event_cancelled'})
        alt success
            RS->>RS: successfulRefunds++, totalRefundAmount += amount
        else failure
            RS->>RS: failedRefunds++, log error
        end
    end
    end
    RS-->>Admin: BatchRefundResult {totalOrders, successfulRefunds, failedRefunds, totalRefundAmount, results[]}
```

5. Cancellation Types

Type	Triggered By	Notes
user_requested	Customer via UI	Standard refund request
event_cancelled	Admin cancels event	Batch refund all paid orders
admin_cancelled	Admin cancels specific order	Manual override

6. Error States

Error Condition	Behavior
Order not found	Transaction aborted, 404 returned
Order not paid	Returns {success:true, refundAmount:0} (no Stripe call)
Stripe refund fails	Transaction aborted, error propagated
Session commit fails	Full rollback — order + CancellationLog both reverted

7. Data Written on Refund

Order update:

```
paymentStatus: 'refunded'
refundId: stripe_refund_id
refundedAt: Date
```

CancellationLog created:

```
orderId, eventId, userId
reason (string)
cancellationType: 'user_requested' | 'event_cancelled' | 'admin_cancelled'
initiatedBy (User ObjectId)
refundAmount (number)
```

8. Stripe Refund Details

- Amount sent to Stripe in **file** (integer): `Math.round(refundAmount * 100)`
- Stripe `refunds.create` called on the original `paymentIntentId`
- Uses the **same Stripe instance** (platform or vendor) that processed the original payment
- Refund ID stored in Order for reconciliation